All-gap moments and Gaussian limits for ballistic-deposition roots

Principia Math In Progress

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MathDB 342282. Full intended theorem. The complete accepted manuscript and current preprint have been read; the refreshed bounded source and claim audit is recorded with this proof.

Task 1: statement and scope

Consider the cyclic strip with K sites. At each step a uniform site j is selected and its height becomes $1 + \max(H(j-1), H(j), H(j+1))$; other heights are unchanged. Initially all heights are zero. A root is a site that ever receives a particle at height one. For $i \geq 1$, $D_{i,K}$ counts consecutive roots at cyclic distance $g = i + 1$, including the wrapping pair.

The first conjecture in Section 3.5 of Mansour–Rastegar–Roitershtein [1] asks for linear moments and a normal limit. It prints $\sigma(D_{i,K}) = b_i K$, although σ denotes standard deviation. The surrounding discussion and Theorems 3.3 and 3.5 concern linear *variance*. We prove that intended all-i assertion and both Gaussian conjectures. The elementary variance/standard-deviation wording discrepancy earns no research credit.

Theorem 1. *For every fixed $i \geq 1$, there are positive rational numbers a_i, b_i such that, whenever $K \geq 2(i+4)$,*

$$\mathbb{E}D_{i,K} = a_i K, \qquad \operatorname{Var}(D_{i,K}) = b_i K. \tag{1}$$

Moreover,

$$\frac{D_{i,K} - a_i K}{\sqrt{b_i K}} \implies N(0,1). \tag{2}$$

The vector of all coordinates standardized by their actual means and standard deviations converges weakly in $\mathbb{R}^{\mathbb{N}}$, with its product topology, to a centered Gaussian process. For the finitely small sizes where a coordinate has zero variance, define that standardized coordinate as zero; also set coordinates $i \geq K$ to zero.

Independent priorities and local gap indicators

Run independent rate-one Poisson clocks at the sites. Their merged sequence of site labels is independent and uniform, so it gives the same deposition process. Its first arrival times are independent exponentials. A root forms at j exactly when its first arrival precedes the first arrivals at both neighbors: otherwise a neighbor already has positive height. Later arrivals cannot create a root at a previously occupied site. Applying the exponential distribution function gives independent uniform priorities U_j and the exact joint representation

$$\xi_j = \mathbf{1}\{U_j < U_{j-1},\ U_j < U_{j+1}\}.$$

All indices here are cyclic. If $K > g + 2$, put

$$Y_j^{(g)} = \xi_j \xi_{j+g} \prod_{t=1}^{g-1} (1 - \xi_{j+t}). \qquad D_{i,K} = \sum_{j=1}^{K} Y_j^{(g)}. \tag{3}$$

The indicator depends only on $U_{j-1}, \ldots, U_{j+g+1}$, a window of length $m = g + 3 = i + 4$.

Use the same definition on a two-sided infinite independent sequence. Its indicators are stationary and $(m-1)$-dependent: disjoint priority windows give independent families. Define

$$a_i = \mathbb{E} Y_0^{(g)}, \qquad b_i = \operatorname{Var}(Y_0^{(g)}) + 2 \sum_{h=1}^{m-1} \operatorname{Cov}(Y_0^{(g)}, Y_h^{(g)}). \tag{4}$$

Each probability in (4) is a finite sum of equally likely strict orders of finitely many continuous priorities. Thus a_i, b_i are rational. A configuration with low values at $0, g$, high external neighbors, and increasing interior values has exactly those two roots in $[0, g]$. It occupies a nonempty open set, so $a_i > 0$.

For $K \geq 2m$, a single window has the line law. A pair of overlapping windows has the line law at offset h, or at offset $-h$ across the cyclic seam. All other pairs are independent. Summing covariances gives (1) with the constants (4), except that positivity of b_i still requires proof.

Strictly positive variance density

Fix $g \geq 2$, and write $q = 2g - 1$, $L = 2g + 3$. Partition $\lfloor K/L \rfloor$ disjoint consecutive blocks of length L, leaving any remaining sites aside. Label sites within each block $-1, 0, 1, \ldots, 2g, 2g + 1$. Its anchors are $0, 2g$, guards are $-1, 2g + 1$, and interior sites are $1, \ldots, q$. A block is good if both anchors have priority below $1/4$, both guards have priority above $3/4$, and all interior priorities exceed $1/4$. Its probability is

$$p = 4^{-4}(3/4)^q > 0. \tag{5}$$

Reveal all anchor, guard, and leftover priorities, and for every block reveal whether all its interior priorities exceed $1/4$. In blocks that are not good, reveal the whole interior as well. Conditional on this information, the interiors of the good blocks are independent; in each good block they are independent uniforms on $(1/4, 1)$. Both anchors are forced roots and both guards are forced nonroots. All roots outside the good interiors are determined.

Consequently $D_{i,K}$, conditionally, is a constant plus the sum of independent quantities H, one per good block, where H counts gaps of length g inside its anchored segment $[0, 2g]$. The fixed anchor roots separate all such gaps from exterior gaps. Two interior order patterns have different values of H. If $U_1 < \cdots < U_q$, there are no interior roots, so $H = 0$. For the second pattern make U_g the least interior priority, make $U_1, \ldots, U_{g-1}$ increase, and make $U_{g+1}, \ldots, U_q$ decrease. Choose one fixed total order with these properties. Its only interior root is g, so $H = 2$. These descriptions also hold when $g = 2$.

Each of the two selected total orders has conditional probability $1/q!$. For an independent conditional copy H',

$$\operatorname{Var}(H \mid \text{revealed}) = \tfrac{1}{2}\mathbb{E}((H - H')^2 \mid \text{revealed}) \geq \frac{4}{(q!)^2}. \tag{6}$$

Conditional independence, total variance, and (5) now give

$$\operatorname{Var}(D_{i,K}) \geq \left\lfloor \frac{K}{L} \right\rfloor \frac{4p}{(q!)^2}.$$

Combining this with (1) and sending K to infinity proves the explicit positive bound

$$b_i \geq \frac{4^{-3}(3/4)^{2g-1}}{(2g+3)((2g-1)!)^2} > 0. \tag{7}$$

The scalar central limit theorem

We include a blocking argument. On the infinite line let $X_j = Y_j^{(g)} - a_i$ and $S_K = \sum_{j=1}^{K} X_j$. Finite dependence gives $\mathrm{Var}(S_K) = Kb_i + O(1)$. Take consecutive long blocks of length $\ell = \lfloor K^{1/4} \rfloor$, separated by $m-1$ discarded indices, and discard the final incomplete block. The long-block sums B_r are independent and centered, and $|B_r| \leq \ell$. The discarded sum R_K has variance

$$\mathrm{Var}(R_K) = O(K/\ell + \ell) = o(K),$$

since each discarded index has at most $2m-2$ dependent neighbors. The L^2 triangle inequality yields $s_K^2 := \sum_r \mathrm{Var}(B_r) \sim b_i K$.

For completeness, the independent-block normal limit follows from the Taylor expansion of characteristic functions. For each fixed real t, the total third-order remainder is bounded by

$$\frac{|t|^3}{6s_K^3} \sum_r \mathbb{E}|B_r|^3 \leq \frac{|t|^3 \ell}{6 s_K} \longrightarrow 0.$$

Also $\max_r \mathrm{Var}(B_r)/s_K^2 \leq \ell^2/s_K^2 \to 0$. Multiplying the characteristic functions, or taking their logarithms near one, therefore gives $\mathbb{E}\exp(it\sum_r B_r/s_K) \to e^{-t^2/2}$. Thus $S_K/\sqrt{b_i K}$ converges to $N(0,1)$.

Couple the cycle to the line by using the same priorities at sites $1, \ldots, K$. Only a bounded number, depending on m, of indicator windows cross the seam. The cyclic and line sums therefore differ by a deterministically bounded amount. Dividing by $\sqrt{b_i K}$ proves (2).

The joint product-space limit

For fixed r, apply the same argument to any real linear combination of the first r local indicators, centered and divided by $\sqrt{b_i}$. This remains bounded and finitely dependent. Its variance density is nonnegative. If positive, the blocking proof applies after rescaling; if zero, its line-sum variance is $O(1)$, so its sum divided by $\sqrt{K}$ converges to zero in L^2. The bounded seam correction still vanishes.

The Cramer–Wold criterion gives a centered Gaussian limit for every finite set of coordinates, with covariance matrix

$$\Sigma_{ij} = \frac{1}{\sqrt{b_i b_j}} \sum_{h \in \mathbb{Z}} \mathrm{Cov}\big(Y_0^{(i+1)}, Y_h^{(j+1)}\big). \tag{8}$$

Each sum is finite. These matrices are consistent and positive semidefinite, as limits of covariance matrices; their diagonals are one. They define a Gaussian law on the countable product.

Every standardized coordinate has second moment at most one. Given $\varepsilon > 0$, choose $M_i > 0$ with $\sum_i M_i^{-2} < \varepsilon$. By Chebyshev and the union bound, all these vector laws give mass at least $1-\varepsilon$ to the compact product $\prod_i [-M_i, M_i]$. They are uniformly tight. Finite-dimensional convergence and tightness prove convergence in the product topology to the Gaussian law with covariance (8).

Tasks 2–5: four proposed mathematical follow-ups

Task 2: Derive closed formulas, or generating functions, for all mean and variance constants a_i, b_i in (4).

Task 3: Prove a quantitative normal approximation and a local central limit theorem for $D_{i,K}$, with explicit dependence on fixed i.

Task 4: Classify Gaussian and Poisson limiting regimes when the gap index $i = i(K)$ grows with the strip size.

Task 5: Determine sufficient conditions on nonuniform site arrival rates for analogous linear-variance and joint Gaussian limits.

These are proposed directions, not additional claimed solutions.

Recorded verification. Four exact groups check 1,093 adjacent-order patterns; evaluate the constants for $1 \le i \le 10$, agreeing with all seven known pairs; check 409,112 first-arrival orders modulo rotation for $3 \le K \le 10$; and check variance witnesses for $2 \le g \le 50$, including exhaustive conditional variances for $g = 2, 3, 4$. These finite computations supplement the proof; they do not establish an infinite-parameter statement or a normal limit by numerical fitting. The proof has been checked only by its authoring agent, without independent peer review or formal proof-assistant verification.

Sources and status

[1] T. Mansour, R. Rastegar, A. Roitershtein.
On ballistic deposition process on a strip.
arXiv:1903.12548v2 (19 June 2019), all 27 PDF pages read. The target is Section 3.5, p. 23; the gap definition is on p. 10.

Journal article: *J. Stat. Phys.* 177 (2019), 626–650, journal DOI. The complete accepted manuscript, PMC6913540 / NIHMS1058046, including Sections 1–3.5, all three tables, references, and all seven figures, was read on 22 September 2026. Its Section 3.5 retains both target conjectures. PMC's version policy identifies this as the manuscript accepted after peer review. The separately typeset publisher version remains unread; no identity with that version is asserted. The accepted final supplies the source-scope check, replacing the earlier pending assessment.

Status at 22 September 2026: complete intended mathematical result; eligible within the recorded bounded audit. Initial and refreshed MathDB and bounded public-claim checks are recorded in the package. No credit is claimed for correcting the printed standard-deviation formula, for known finite-i moment calculations, or separately for the joint strengthening.